



Patient Profile

Name: DUMMY

Age:

Gender:

Summary of Results

The predictive model classifies this laboratory profile as within normal limits. However, individual test values reveal notable deviations that merit clinical attention.

Detailed Test Explanations

AST (SGOT) – An aminotransferase primarily found in liver and cardiac muscle. Levels below the lower reference limit (<10 U/L) suggest reduced hepatic cell turnover or potential assay interference; values above the upper limit typically indicate hepatocellular injury. The patient's result is below the reference range.

ALT (SGPT) – A liver-specific aminotransferase; elevations suggest hepatocellular damage, whereas low values (<10 U/L) can reflect low muscle mass or assay variability. This result is below the normal range.

AST:ALT Ratio – A ratio of approximately 1.00, within the expected value for non-pathologic conditions. A ratio significantly above 1 may signal alcoholic liver disease or other hepatic insults.

GGTP – A marker of biliary tract health and alcohol consumption. Values below 10 U/L indicate no cholestatic or biliary pathology.

Alkaline Phosphatase (ALP) – Enzyme related to bone and biliary epithelium; levels under 30 U/L are below the reference range and may suggest low bone turnover, malnutrition, or hypoparathyroidism.

Bilirubin Total – 0.90 mg/dL falls comfortably within the normal 0.30–1.20 mg/dL range, indicating adequate hepatic conjugation and excretion.

Bilirubin Direct – 0.20 mg/dL is within the normal <0.30 mg/dL range, supporting normal hepatic processing.

Bilirubin Indirect – 0.70 mg/dL is within the normal <1.10 mg/dL range.

Total Protein – >12.00 g/dL far exceeds the normal 5.70–8.20 g/dL range, indicating hyperproteinemia.

Albumin – 2.00 g/dL is markedly below the reference 3.20–4.80 g/dL, signifying hypoalbuminemia.

A:G Ratio – 0.10 is well below the expected 0.90–2.00, reflecting a disproportionate increase in globulin relative to albumin.

Overall Interpretation

The liver enzyme panel (AST, ALT, GGTP, ALP, bilirubin) remains largely unremarkable, supporting normal hepatic function. Contrastingly, the markedly elevated total protein with concurrent hypoalbuminemia and a severely depressed A:G ratio point to a substantial rise in globulins. This pattern is characteristic of chronic immune stimulation, infection, or monoclonal gammopathy such as multiple myeloma. Despite these biochemical abnormalities, the predictive model's

overall assessment remains “Normal,” likely because the liver-specific indices are within acceptable limits. Nevertheless, the protein derangement warrants further investigative evaluation.

Recommendations

1. **Specialist Referral** – Discuss findings with a hematologist or an oncologist to evaluate for plasma-cell dyscrasia or other causes of hyperglobulinemia.
2. **Serum Protein Electrophoresis (SPEP)** – To detect monoclonal protein spikes and differentiate between polyclonal and monoclonal gammopathy.
3. **Immunofixation and Quantitative Immunoglobulin Levels** – To delineate specific Ig classes contributing to the high globulin load.
4. **Comprehensive Metabolic Panel** – Reassess liver function and renal status to rule out other contributing factors to protein abnormalities.
5. **Nutritional Assessment** – Evaluate dietary protein intake and possible malabsorption contributing to hypoalbuminemia.
6. **Follow-up Testing** – Repeat liver enzymes and protein profile in 4–6 weeks to monitor trends.
7. **Lifestyle Measures** – Maintain adequate hydration, balanced protein intake, and avoid hepatotoxic substances.

Disclaimer

This report is informational and not a substitute for professional medical diagnosis or treatment. Always consult a qualified healthcare provider for personal medical advice.